MIL-T-20215C
3 December 1985
SUPERSEDING
MIL-B-20215B(NAVY)
24 August 1964
(See 6.3)

MILITARY SPECIFICATION

TILE REFRACTORY, SILICON CARBIDE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers silicon carbide refractory tile for use as baffles in naval boiler furnaces.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 9350

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2.1.2 Other Government drawing. The following other Government drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
NAVSHIPS 810-1385936 - Tile, Baffle, Silicon Carbide for Marine Boilers.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
C 20 - Test Methods for Apparent Porosity, Water Absorption, Apparent Specific Gravity, and Bulk Density of Burned Refractory Brick and Shapes by Boiling Water.
C 571 - Standard Methods for Chemical Analysis of Carbon and Carbon-Ceramic Refractories.
D 3951 - Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The tile shall be composed of a bonded silicon carbide material formed to a uniform, dense structure.

3.2 Density. The bulk density shall be not less than 2.50 grams per cubic centimeter (see 4.4.1).

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3.3 Porosity. The apparent porosity shall be not more than 18 percent (see 4.4.1).

3.4 Chemical composition. The silicon carbide content shall be not less than 85 percent by weight of the finished product. The product shall contain not more than 0.5 percent of lime (CaO), not more than 0.25 percent of alkalies, and not more than 0.1 percent of magnesia (MgO) (see 4.4.2).

3.5 Dimensions. Tile shall be furnished in the size and shape specified (see 6.2), and shall conform to the dimensions and tolerances shown on Drawing 810-1385936.

3.6 Marking. Each tile shall be marked with the piece number and the manufacturer's brand name or symbol in accordance with Drawing 810-1385936.

3.7 Workmanship. The silicon carbide tile shall be of uniform structure, density and porosity throughout with no cracks, voids or soft centers. The tile shall have smooth dense surfaces. Edges and corners shall be sufficiently solid and strong to prevent crumbling and chipping during normal handling.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.3 Inspection of end item.

4.3.1 Examination of the end item. Examination of the end item shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) (see table I). The inspection levels for determining the sample size, and the AQLs expressed in defects per 100 units are given in table I.

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TABLE I. Classification of defects.

Examine	Defect	Classification		Inspection level	AQL
		Major	Minor		
Workmanship	Cracked, broken laminations	X		S-4	2.5
	Surfaces, not smooth or dense	X		S-4	2.5
	Corner or edges chipped or crumbled affecting service-ability	X		S-4	2.5
Dimensions	Any specified dimension not within the tolerances specified on Drawing 810-1385936	X		S-4	2.5
Packaging	Packaging material not as specified	X		S-1	4.0
	More than one size or shape per package	X		S-1	4.0
	No level specified	X		S-1	4.0
	Container not as specified, closures not accomplished by specified or required methods or materials	X		S-1	4.0
	Any nonconforming component, component missing, damaged or otherwise defective, affecting serviceability	X		S-1	4.0
	Inadequate application of component, such as incomplete closures of container flaps, loose or inadequate strap-pings, bulged or distorted containers		X	S-1	4.0
Count	Number of tiles per container less than indicated quantity		X	S-1	4.0
Weight	Gross weight exceeds speci-fied requirements		X	S-1	4.0
Markings	Omitted, illegible, incorrect, incomplete	X		S-1	4.0

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4.3.2 Testing of the end item. Samples of the tile selected from each lot presented for examination shall be tested for bulk density, apparent specific gravity and chemical analysis.

4.3.2.1 Bulk density and apparent specific gravity. The inspection level for determining the sample size shall be S-3. The acceptable quality level expressed in defects per 100 units shall be 6.5.

4.3.2.2 Chemical analysis. The sample size for chemical analysis shall be 5 tiles, selected at random. If the average chemical composition (see 4.4.2) does not conform to the requirements of 3.4, this shall be cause for rejection of the lot represented.

4.4 Test procedures.

4.4.1 Bulk density and apparent porosity. The bulk density and apparent porosity shall be determined in accordance with ASTM C 20. Samples shall be tested intact without cutting or breaking.

4.4.2 Chemical analysis.

4.4.2.1 Sample preparation. A single piece, weighing approximately 4-6 grams and free of original surface or skin, shall be broken from each of the five sample tiles. Each piece so selected shall be crushed by impact, preferably with a pneumatic hammer in a tool steel mortar, and not by grinding, sufficiently to pass through a 100-mesh standard screen. Samples prepared in this manner from each sample unit shall be combined, thoroughly mixed and quartered. From the composite sample, duplicate samples shall be taken for two parallel chemical analyses.

4.4.2.2 Chemical analysis procedure. Chemical analysis, to establish compliance with 3.4, shall be determined in accordance with ASTM C 571. The average of the two determinations for each constituent shall be reported.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition).

5.1 Preservation. Preservation shall be level A, C or commercial, as specified (see 6.2). Packs shall contain tile of one size and shape only.

5.1.1 Level A. Tile shall be packed in fiberboard boxes conforming to PPP-B-636 weather-resistant. Boxes shall be of variety DW (double wall) or variety SW (single wall). SW boxes shall be provided with a full height inside liner of the same grade as the box used. Tiles shall be individually wrapped with cushioning media or segregated through the use of fiberboard partitions. Additional cushioning as required shall be used to fill all void spaces within the boxes. Loose fill material including straw or sawdust shall not be used. Boxes shall be closed and reinforced with non-metallic tape or banding with box-closure conforming to method V of the appendix to the box specification. Gross weight of boxes shall not exceed 40 pounds.

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5.1.2 Level C. Tile shall be packed as specified in 5.1.1 except that boxes shall be of the domestic class, minimum grade 200 and box closure shall conform to method I of the appendix to the box specification.

5.1.3 Commercial. Tile shall be packed in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, C or commercial, as specified (see 6.2). Unless otherwise specified (see 6.2), shipping containers or pallets shall contain packs of tile of one size and shape only.

5.2.1 Level A. Tile, preserved as specified in 5.1 shall be packed in wood or wood cleated plywood boxes or unitized on pallets conforming to PPP-B-621, class 2, PPP-B-601 overseas type or MIL-STD-147, load type 1a, bonding means "O" or "P" respectively. Box closure and reinforcement shall be in accordance with the box specification or appendix thereto. Gross weight of wood boxes shall not exceed 200 pounds of palletized loads 3000 pounds.

5.2.2 Level B. Tile, preserved as specified in 5.1 shall be packed as specified in 5.2.1 except that wood and wood cleated plywood boxes may be of the domestic type or class.

5.2.3 Level C. Tile, preserved as specified in 5.1 may be shipped without overpacking. When specified (see 6.2) preserved tiles shall be in unit loads on pallets conforming to MIL-STD-147 with load type and bonding means at the option of the contractor.

5.2.4 Commercial. Tile, preserved as specified in 5.1 shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required, (see 6.2) level A, B and C interior and exterior boxes and palletized unit loads shall be marked in accordance with MIL-STD-129 include bar code markings. Commercial interior and exterior boxes and palletized unit loads shall be marked in accordance with ASTM D 3951 and shall include bar code marking of MIL-STD-129.

6. NOTES

6.1 Intended use. The silicon carbide refractory tile is intended for use as baffles in naval boiler furnaces.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size and shape of tile required (see 3.5).
(c) Level of preservation and packing required (see 5.1 and 5.2).
(d) When shipping containers or pallets may contain other than tile of one size and shape only (see 5.2).
(e) Special marking required (see 5.3).

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6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Navy - SH
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Review activity:
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User activities:
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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-20215C	TILE REFRACTORY, SILICON CARBIDE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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